

Ilari Angervuori

Satellite Systems Researcher (R&D) | Stochastic Analysis and Simulation

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[GitHub](#), [Google Scholar](#), [LinkedIn](#)

Profile

R&D scientist in non-terrestrial networks (NTN) with a pioneering publication record in stochastic geometry modelling of satellite communication systems, including a first-author publication in the high-impact journal IEEE Transactions on Communications (IEEE TCOM) and a first-author presentation at the flagship conference IEEE GLOBECOM. A mindset that seeks simplicity – not complexity. Enjoys applying existing knowledge in novel environments. Committed to bringing my unique expertise to a collaborative, development-driven, fast-paced environment at a company that is intertwined with ethical questions and AI. Always striving to grow professionally and spiritually, and to cultivate compassion and humility in all aspects of life. Available to relocate; FI/EU citizen.

Skills

Soft:

- Data analysis and visualisation
- Inter-institute collaboration
- Writing
- Research
- Stakeholder management

Hard:

- RF link modelling in satellite communications
- Optimisation and Monte Carlo simulations of large constellations
- Analytical simplifications and full-fidelity validation
- Thermal transfer modelling (heat equation and finite element method (FEM))
- MATLAB, PYTHON, JAVA, C, C++

Spoken Languages:

- Finnish (native)
- English (fluent)
- Swedish (limited working proficiency)
- Spanish (Basics, no working proficiency)

Experience and Impact

5/2025- **Senior Researcher**, Aalto University, School of Electrical Engineering, Espoo, Finland

- Managed 2 R&D projects on mathematical modelling of satellite communications as the corresponding author and handled the full delivery pipeline (simulation, analysis, documentation, peer review) to publication.
- Put forth a closed-form autocorrelation function for aggregate radio frequency (RF) interference at a narrow-beam low Earth orbit (LEO) satellite terminal. Leveraged tractable spectrum analysis and hypothesis classes for artificial intelligence/machine learning (AI/ML) signal processing algorithms, with applications in interference control and medium access control (MAC) design in satellite communications.
- Proposed and analysed successive interference cancellation (SIC) algorithms for multibeam LEO uplink, improving coverage performance by 900% for the third-strongest Earth transmitter in terms of signal-to-interference ratio (SIR). The SIC improves user fairness and enables serving multiple terrestrial terminals per cell, while preserving the average total network throughput at the LEO satellites, which is optimised around 1 (bit/s/Hz). As a trade-off, throughput slightly decreases compared to a one-per-cell link. The mathematical analysis was tested and validated against more realistic Monte Carlo simulations. The system-level insights imply design recommendations for next-generation LEO networks.
- Peer-reviewed journals (including top-impact), IEEE TCOM, IEEE TWC, IEEE TMC, etc.

8/2023- **Research Internship**, University of Notre Dame, Dept. of Electrical Engineering, South Bend, IN, US

- 2/2024 • Secured project financing from the Väisälä Foundation for an international satellite communications R&D project in collaboration with leading experts in stochastic geometry and wireless communications.
- Coordinated an internationally distributed team and handled the full delivery pipeline (simulation, analysis, documentation, peer review) to publication in the high-impact journal IEEE TCOM.
- Built a novel system model and mathematical analysis of link budget and RF interference in a LEO uplink, characterising the SIR using a Lomax distribution. The analysis was tested and validated against more realistic Monte Carlo simulations. Consequently, Lomax distribution results can be utilised for efficient performance analysis and optimisation of LEO communications, providing insight and serving as a hypothesis class in AI/ML algorithms. The results were published in a high-impact journal, IEEE TCOM.
- Identified an optimal cell size in a LEO uplink that maximises average throughput and quantified a trade-off between average throughput and performance consistency: -10 dB cell size provides high average throughput but highly variable link quality of service (QoS) across the beams, whereas -3 dB cells provide more stable QoS but are detrimental to throughput. The results were published in a high-impact journal, IEEE TCOM.
- Mentored junior researchers as a bachelor's degree supervisor and helped establish future talents in wireless communications.

- 09/2020- **PhD Researcher**, *Aalto University*, Signal Processing of Wireless Networks, Espoo, Finland
- 05/2025
- Built, maintained, and documented a multi-language simulation stack (GIT, MATLAB, MATHEMATICA, OCTAVE, PYTHON) including algorithms for analysis, simulation, and visualisation, used across several research projects.
 - Developed a novel, well-cited stochastic geometry framework for a LEO RF uplink that reduced complexity and simulation time compared to MATLAB Satellite Communication Toolbox full-fidelity simulations while producing similar SIR/throughput performance results (statistically relevant results were obtained from initialization to computation in minutes rather than hours). The project was a collaboration with the Technical Research Centre of Finland (VTT).
 - Put forth model simplifications that allow the LEO uplink RF interference to be characterised by a gamma distribution. The model was tested and validated against more realistic Monte Carlo simulations. Leveraged well-known gamma distribution properties for efficient performance analysis and optimisation, insight, and as a hypothesis class for AI/ML algorithms. Presented the paper at the top-tier conference IEEE GLOBECOM in Rio De Janeiro.
 - Proposed, analysed, and simulated a minimal-overhead inter-satellite link (ISL) data-routing algorithm in large constellations using graph algorithms. Optimised a grant-free MAC to maximise throughput and minimise latency.

Education

- Expected **Doctor of Science, PhD**, *Aalto University, School of Electrical Engineering*, Signal processing for communications, • Stochastic geometry analysis and simulation of satellite communications
- 2026 **MSc**, *Univ. of Helsinki*, Applied analysis, • FEM • Partial differential equations (PDEs) • Heat equation
- BSc**, *Univ. of Helsinki*, Applied analysis, • Optimal control theory • System performance optimisation